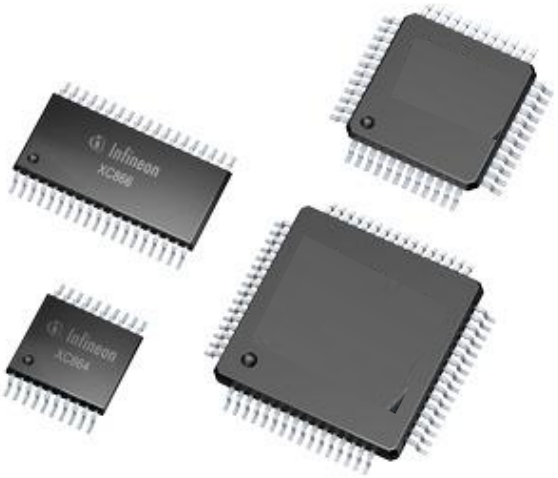
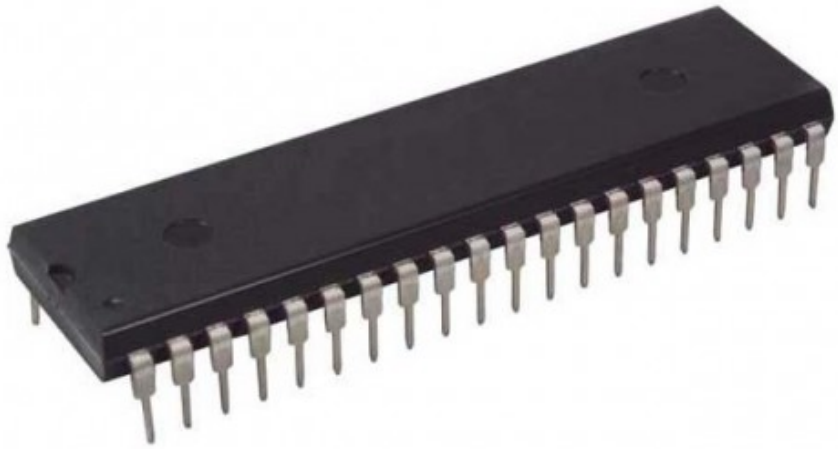
8051 Microcontroller



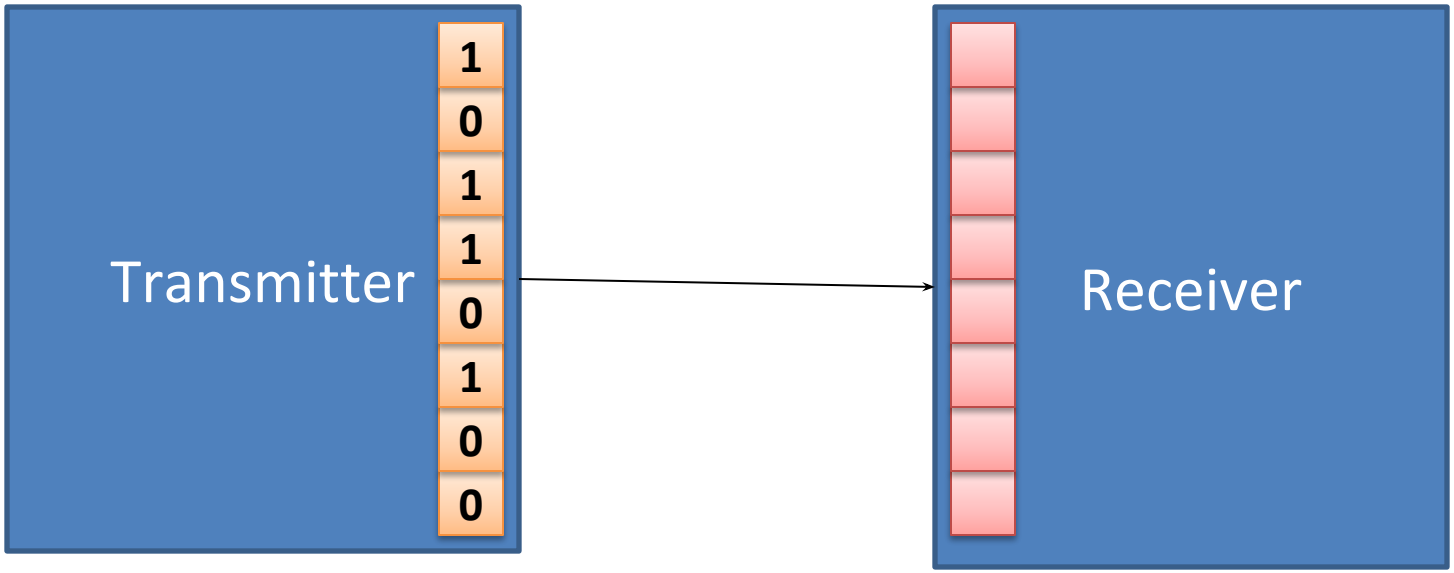


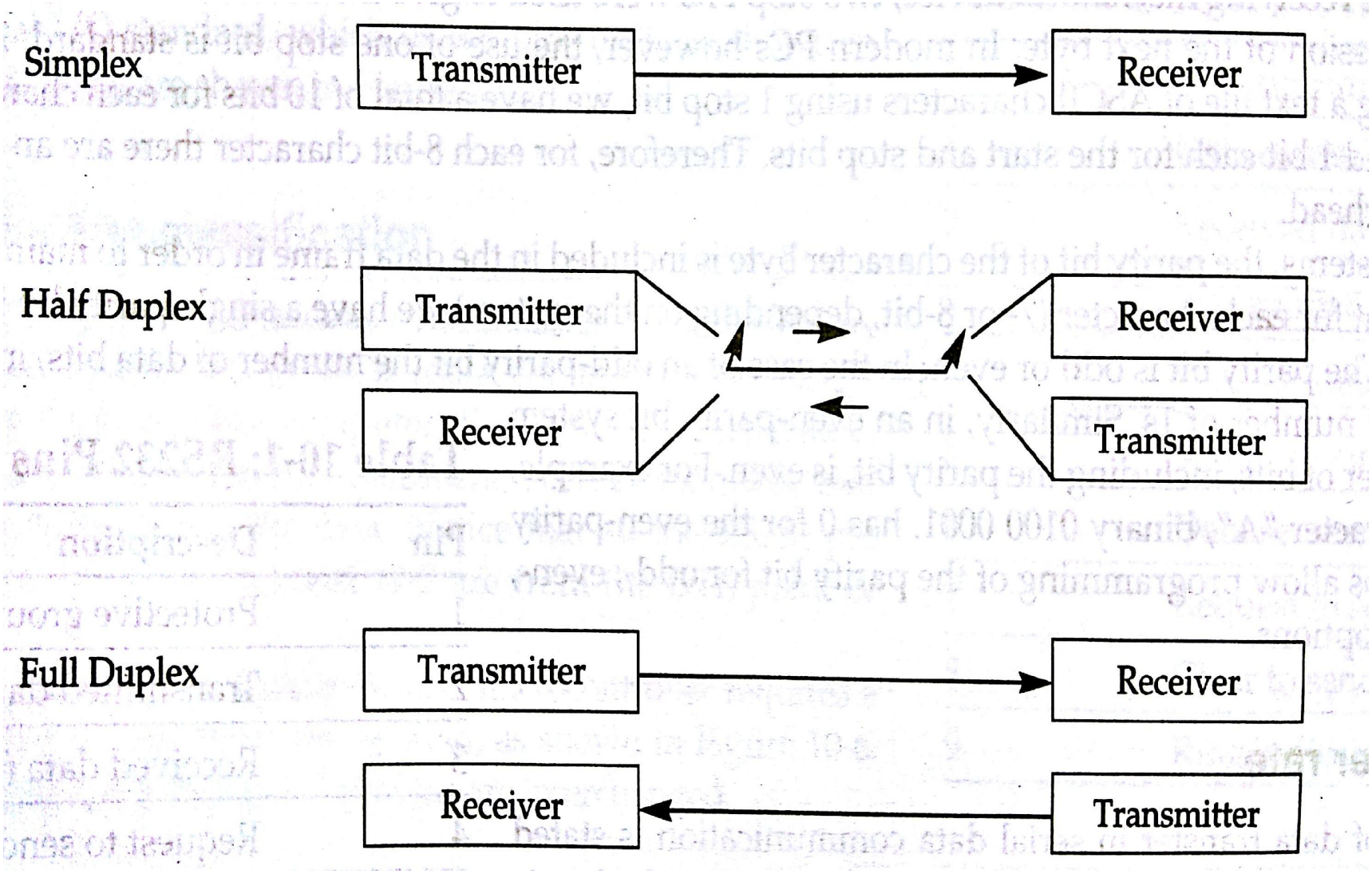
Serial communication in 8051

Serial transfer



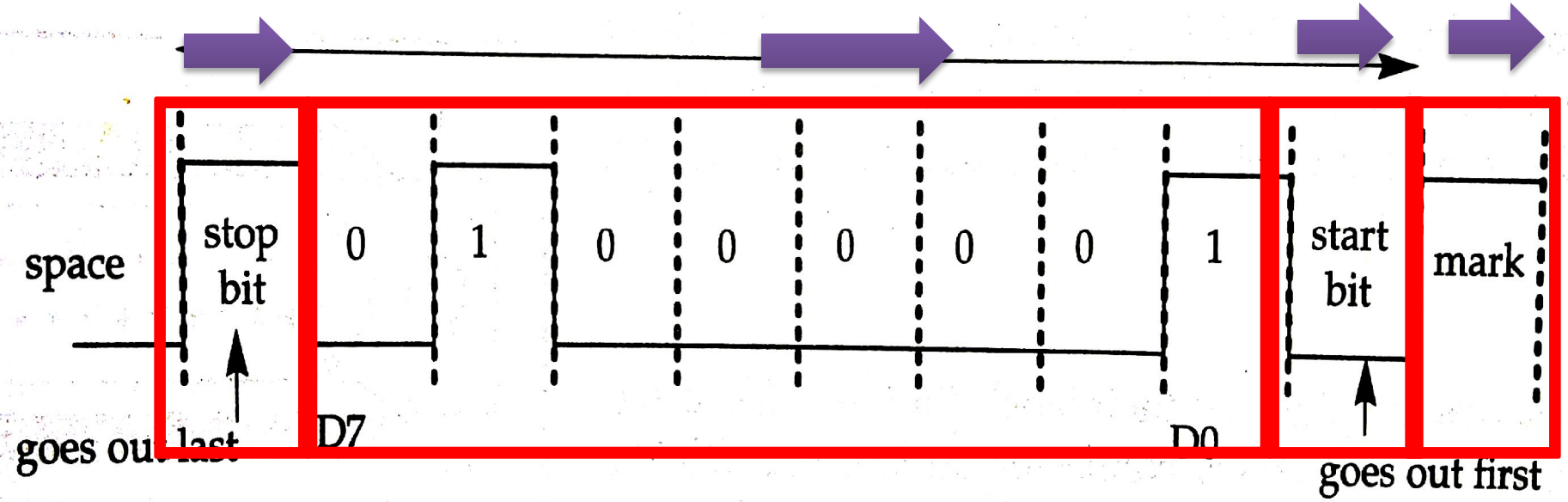
Parallel transfer




Simplex
Transmitter
Receiver
Half Duplex
Transmitter
Receiver
Receiver
Transmitter
Full Duplex
Transmitter
Receiver
Receiver
Transmitter

Asynchronous serial communication

- In asynchronous method, each data is placed between start and stop bits, this is called framing
- The start bit is always 0 (low) and stop bit is 1(high)
- LSB is sent out first
- When there is no transfer, signal is 1 (high), which referred to as mask



Data transfer rate

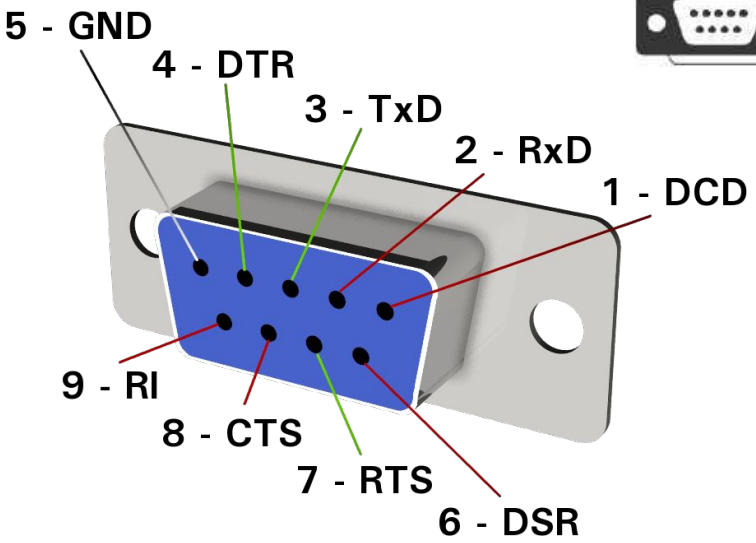
- **Bps:** bit per second
- **Baud rate:** number of signal changes per second

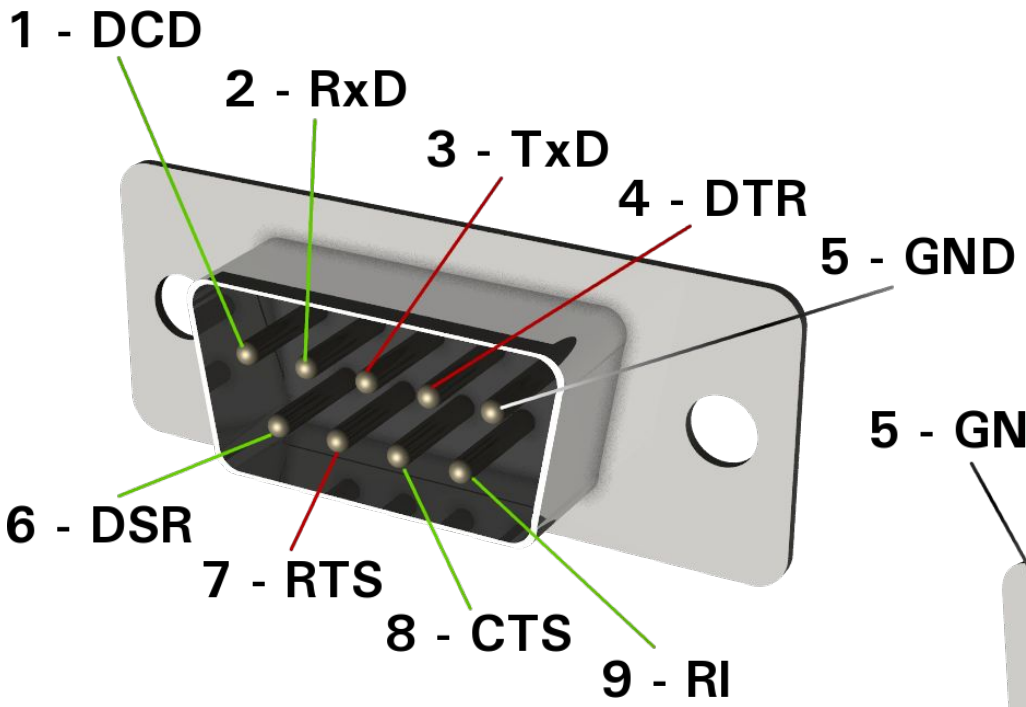
- In case of 8051 bps=baud rate
- RS232 is the most widely used I/O interfacing standard to allow compatibility among data communication equipment.

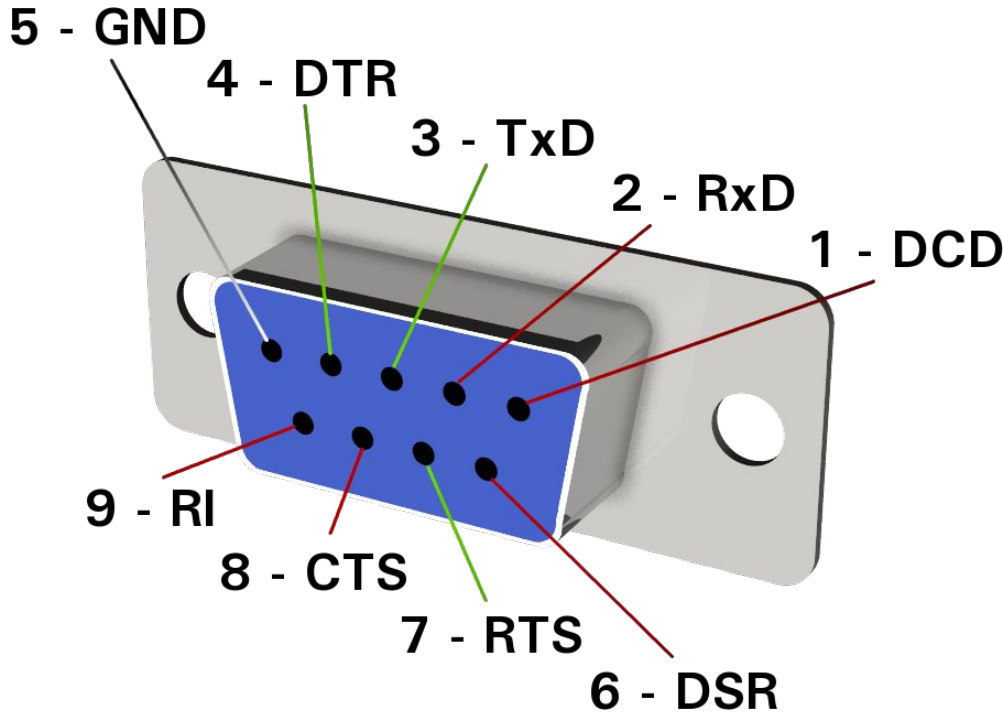
RS232		Microcontroller (TTL)	
Logic 0	Logic 1	Logic 0	Logic 1
3v to 25v	-3v to -25v	0 to 0.8 v	2v to 5v

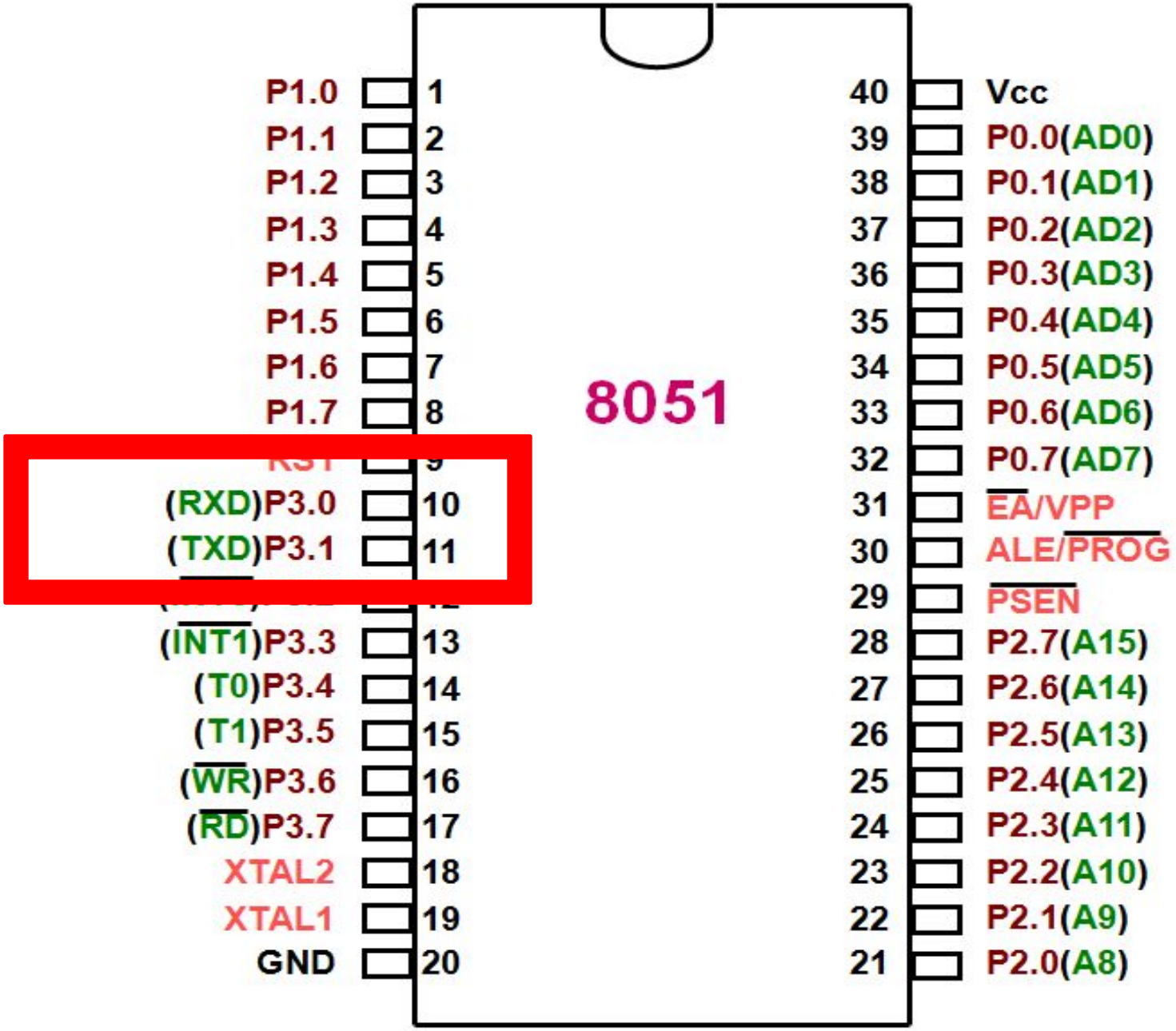

Serie
Rs232
Usb 2.0



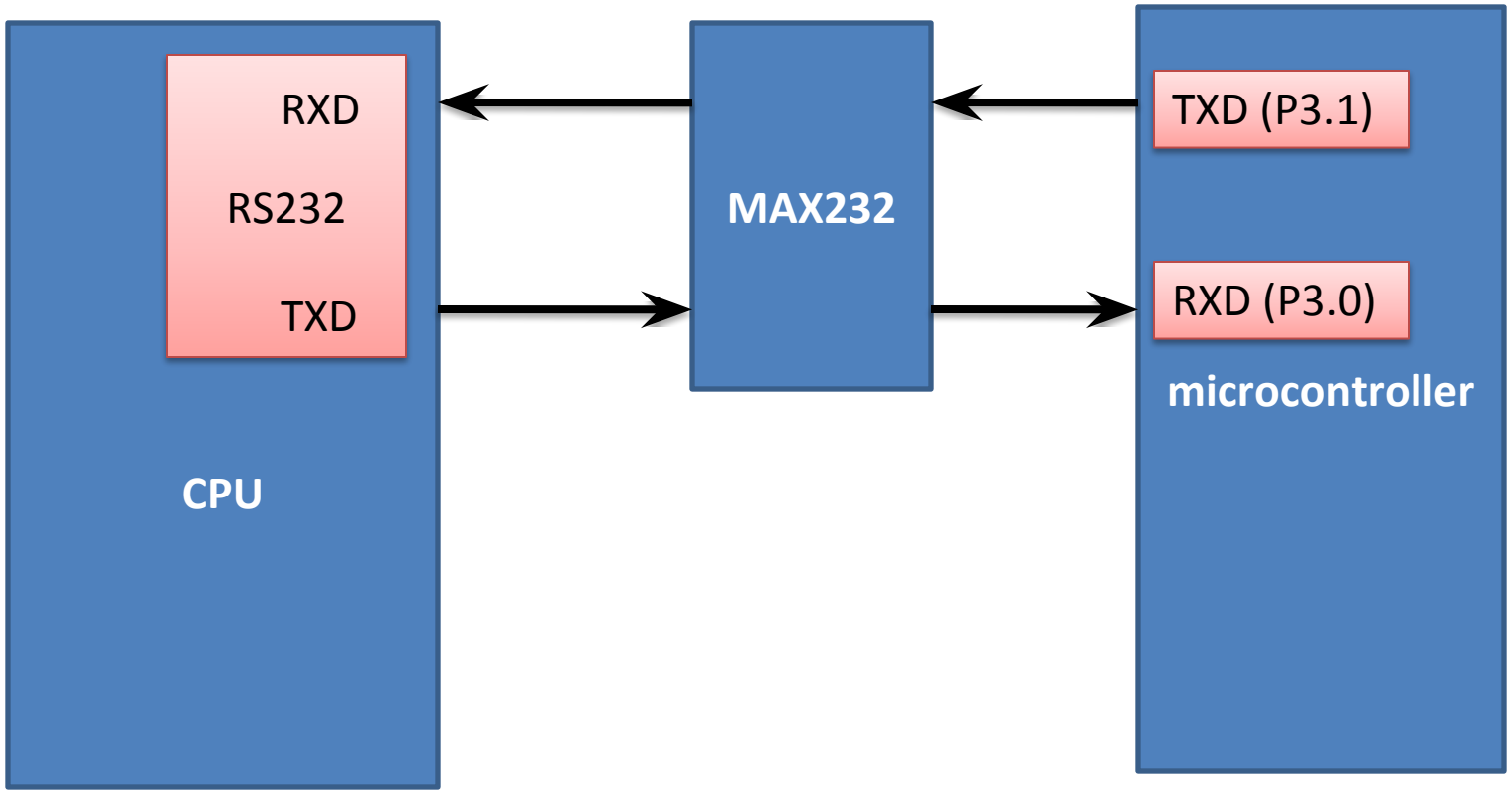

5 - GND
4 - DTR
3 - TxD
2 - RxD
1 - DCD
9 - RI
8 - CTS
7 - RTS
6 - DSR


1 - DCD
2 - RxD
3 - TxD
4 - DTR
5 - GND
6 - DSR
7 - RTS
8 - CTS
9 - RI


5 - GND
4 - DTR
3 - TxD
2 - RxD
1 - DCD
9 - RI
8 - CTS
7 - RTS
6 - DSR


8051
P1.0 1
P1.1 2
P1.2 3
P1.3 4
P1.4 5
P1.5 6
P1.6 7
P1.7 8
(RXD)P3.0 10
(TXD)P3.1 11
(INT1)P3.3 13
(T0)P3.4 14
(T1)P3.5 15
(WR)P3.6 16
(RD)P3.7 17
XTAL2 18
XTAL1 19
GND 20
40 Vcc
39 P0.0(AD0)
38 P0.1(AD1)
37 P0.2(AD2)
36 P0.3(AD3)
35 P0.4(AD4)
34 P0.5(AD5)
33 P0.6(AD6)
32 P0.7(AD7)
31 EA/VPP
30 ALE/PROG
29 PSEN
28 P2.7(A15)
27 P2.6(A14)
26 P2.5(A13)
25 P2.4(A12)
24 P2.3(A11)
23 P2.2(A10)
22 P2.1(A9)
21 P2.0(A8)

MAX232 IC: voltage converter used to convert RS232 signals to TTL voltage levels that will be acceptable to 8051 and vice versa.



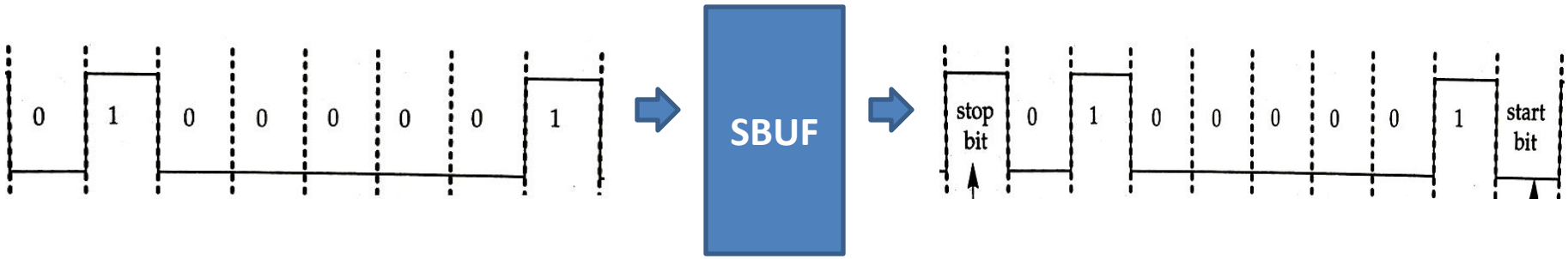
- To allow data transfer between PC and 8051 without any error, we must make sure that baud rate of 8051 matches baud rate of PC's COM port.

- Some of the baud rate supported by PC
  - Baud rate in 8051 is programmable with the help of **timer 1 In mode 2 (8 bit auto reload mode)**

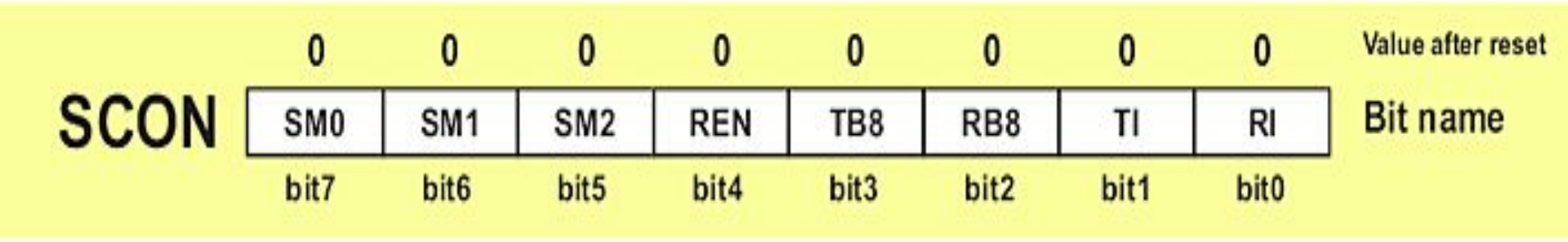
110
150
300
600
1200
2400
4800
9600
19200

SBUF register

- Is 8 bit register
- Byte of data to be transferred via TXD line, it must be placed in SBUF register.
- Similarly SBUF holds the byte of data when it received by the 8051's RXD line
- The moment data is written into SBUF, it is framed with start bit and stop bit and transferred serially via TXD pin



SCON register



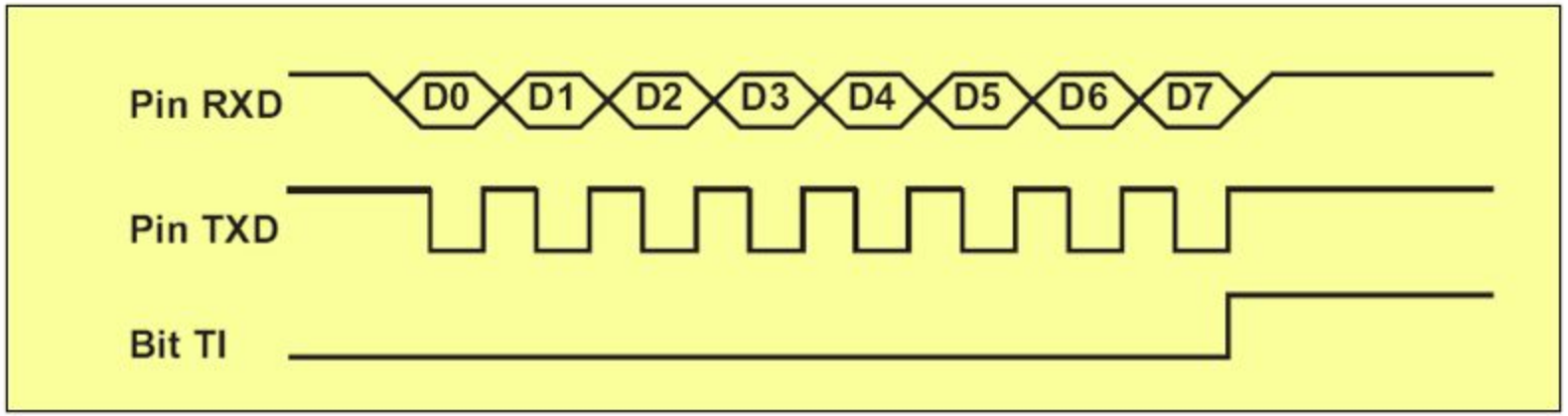
- **SM0** - Serial port mode bit 0 is used for serial port mode selection.
- **SM1** - Serial port mode bit 1.
- **SM2** - Serial port mode 2 bit, also known as multiprocessor communication enable bit.
- When set, it enables multiprocessor communication in mode 2 and 3, and eventually mode 1. It should be cleared in mode 0.

SM0	SM1	MODE	DESCRIPTION	BAUD RATE
0	0	0	8-bit Shift Register	1/12 the quartz frequency
0	1	1	8-bit UART	Determined by the timer 1
1	0	2	9-bit UART	1/32 the quartz frequency (1/64 the quartz frequency)
1	1	3	9-bit UART	Determined by the timer 1

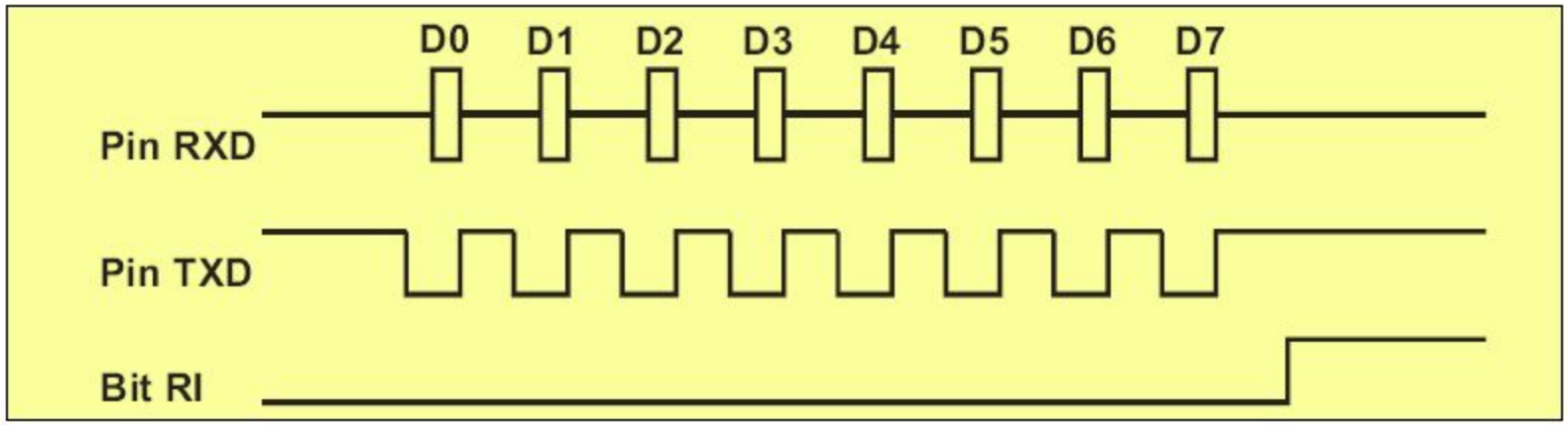
Mode 0

serial data are transmitted and received through the RXD pin, while the **TXD pin output clocks.** The boud rate is fixed at 1/12 the oscillator frequency. On transmit, the least significant bit (LSB bit) is sent/received first.

TRANSMIT



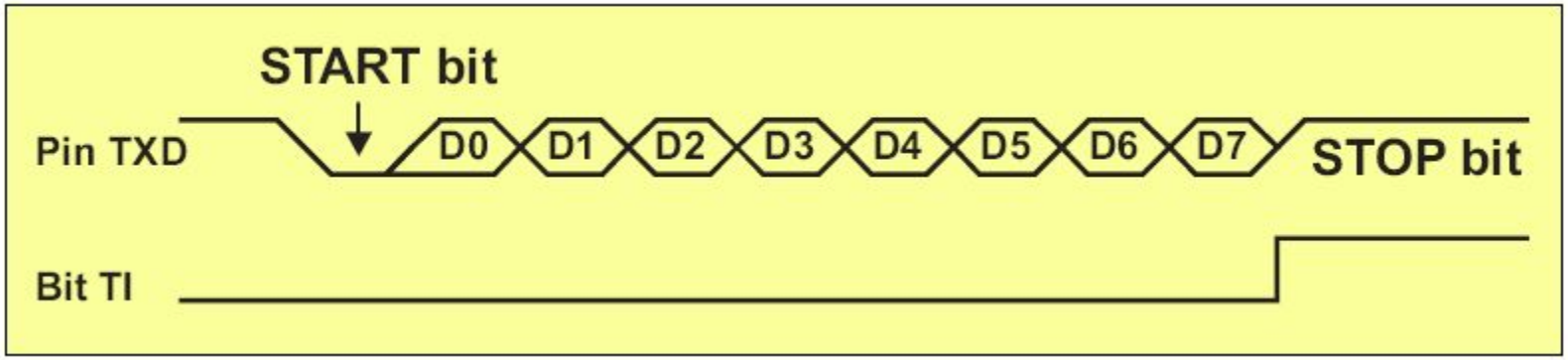
RECEIVE



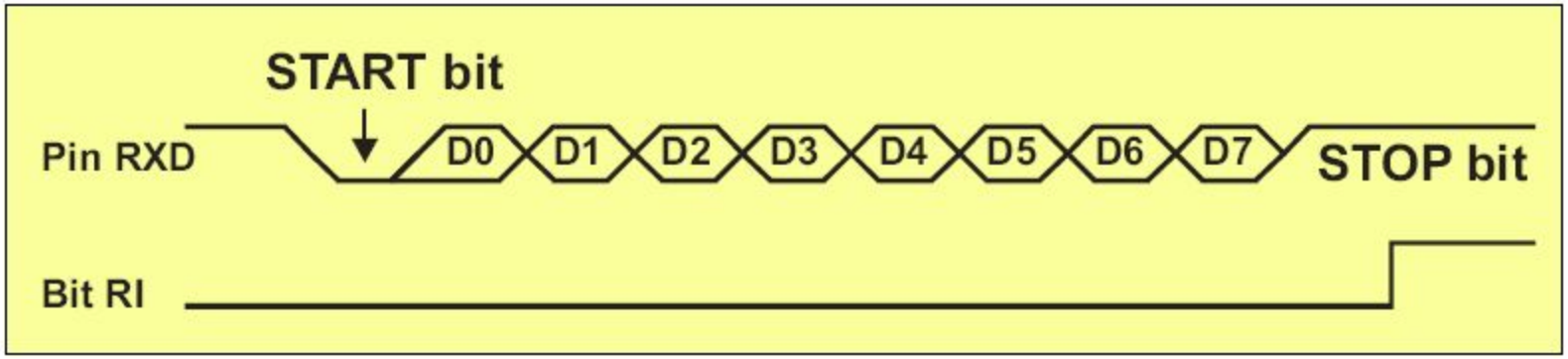
Mode 1

In mode 1, 10 bits are transmitted through the TXD pin or received through the RXD pin in the following manner: a START bit (always 0), 8 data bits (LSB first) and a STOP bit (always 1). The START bit is only used to initiate data receive, while the STOP bit is automatically written to the RB8 bit of the SCON register.

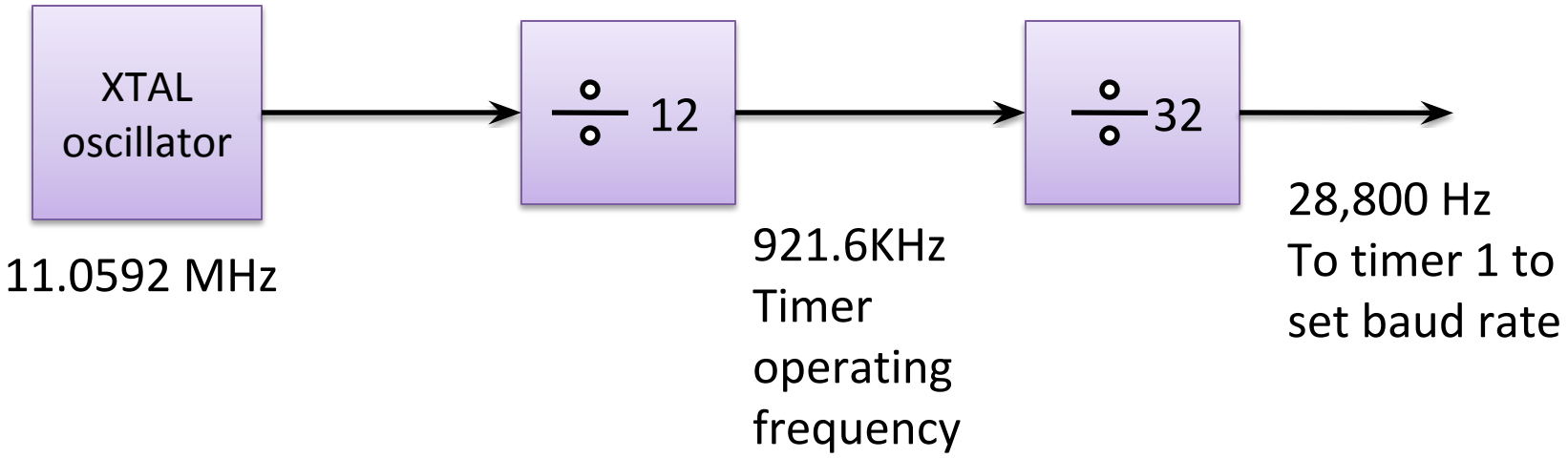
TRANSMIT



RECEIVE



Relation between crystal frequency and baud rate

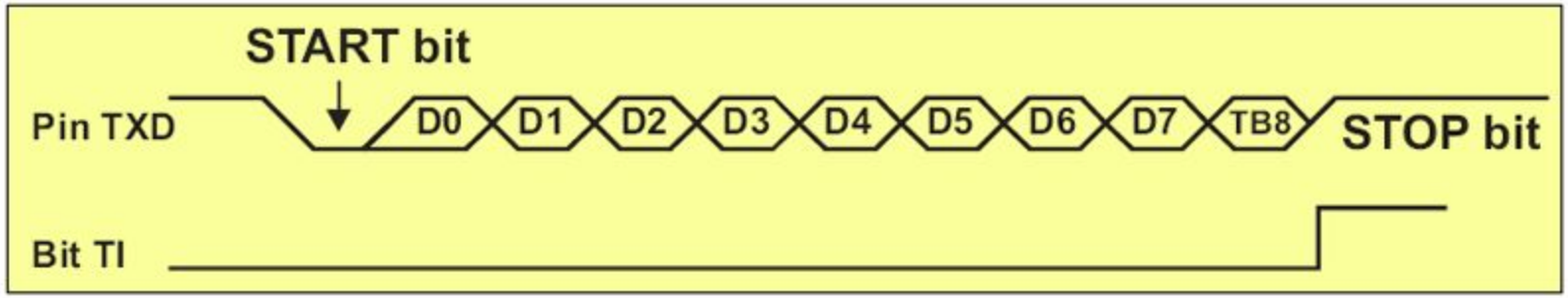


$$TH1 = \left(256 - \left(\frac{28{,}800}{\text{Baud rate}} \right) \right) H$$

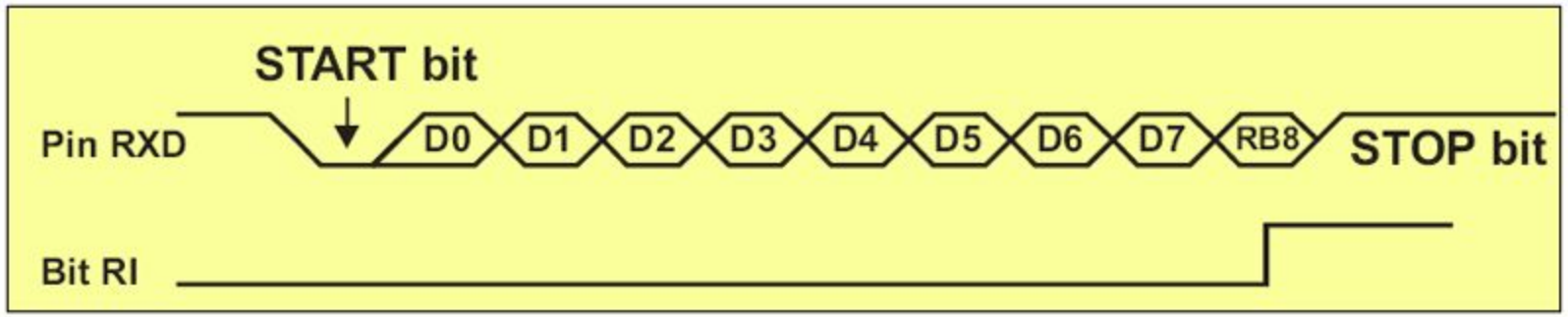
Mode 2

In mode 2, 11 bits are transmitted through the TXD pin or received through the RXD pin: a START bit (always 0), 8 data bits (LSB first), a programmable 9th data bit and a STOP bit (always 1). On transmit, the 9th data bit is actually the TB8 bit of the SCON register. This bit usually has a function of **parity bit.** On receive, the 9th data bit goes into the RB8 bit of the same register (SCON).The baud rate is either 1/32 or 1/64 the oscillator frequency.

TRANSMIT



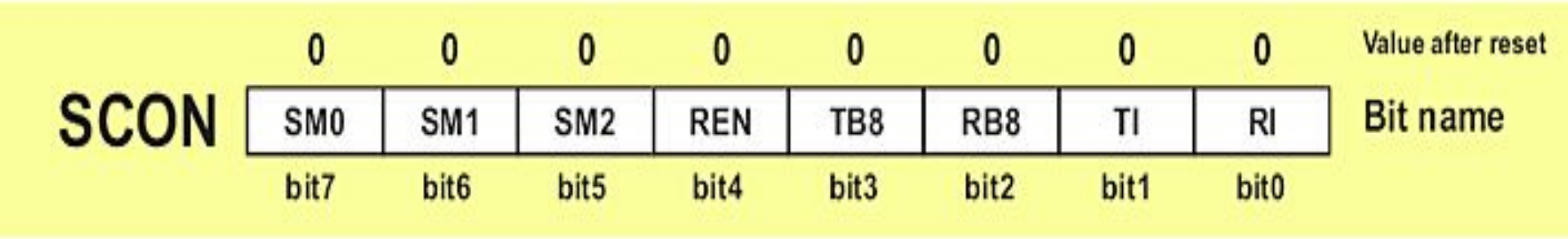
RECEIVE



Mode 3

Mode 3 is the same as Mode 2 in all respects except the baud rate. The baud rate in Mode 3 is variable.

SCON register

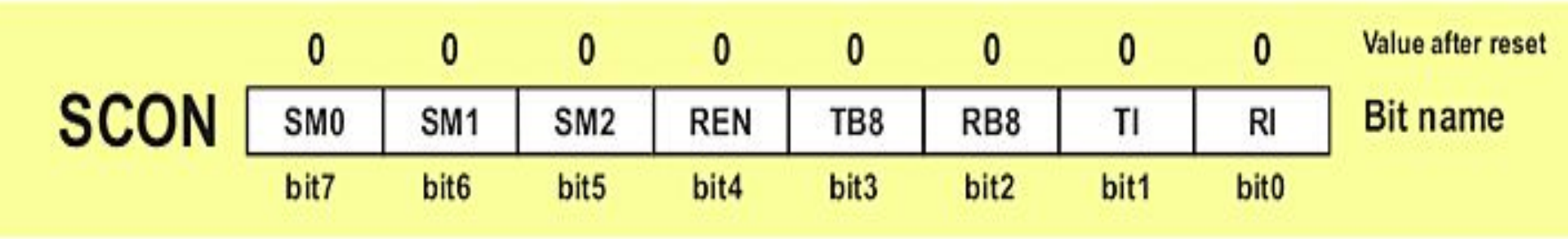


•**REN** - Reception Enable bit enables serial reception when set. When cleared, serial reception is disabled.

•**TB8** - Transmitter bit 8. Since all registers are 8-bit wide, this bit solves the problem of transmitting the 9th bit in modes 2 and 3. It is set to transmit a logic 1 in the 9th bit.

•**RB8** - Receiver bit 8 or the 9th bit received in modes 2 and 3. Cleared by hardware if 9th bit received is a logic 0. Set by hardware if 9th bit received is a logic 1.

SCON register



•**TI** - Transmit Interrupt flag is automatically set at the moment the last bit of one byte is sent. It's a signal to the processor that the line is available for a new byte transmitted. It must be cleared from within the software.

•**RI** - Receive Interrupt flag is automatically set upon one byte receive. it signals that byte is received and should be read quickly prior to being replaced by a new data. This bit is also cleared from within the software.

```
ORG 00H
[redacted]                    ; timer1 mode2
                              ; mode1
                              4800 baud rate
SETB TR1
[redacted]
AGAIN:SJMP AGAIN
END
```